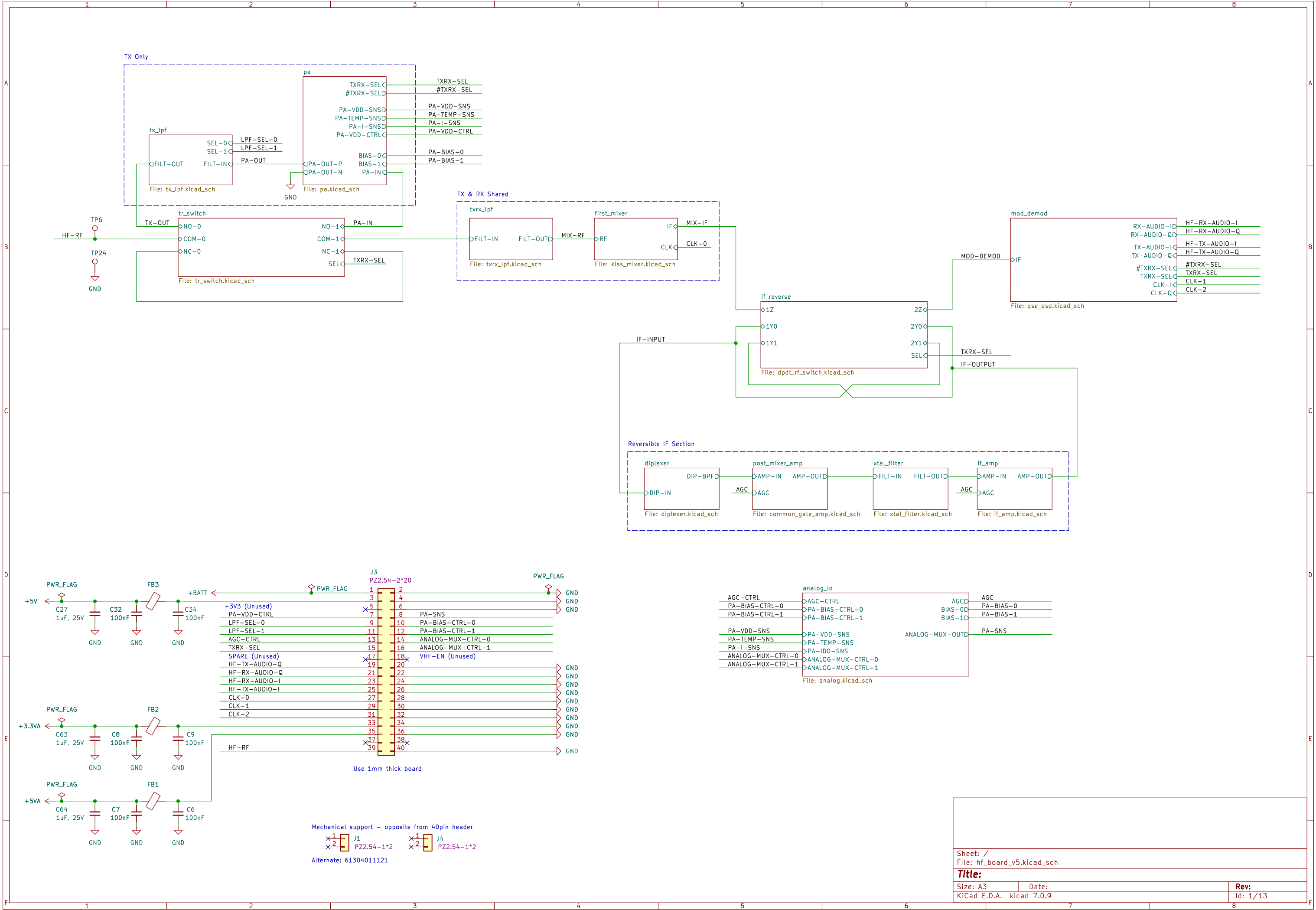
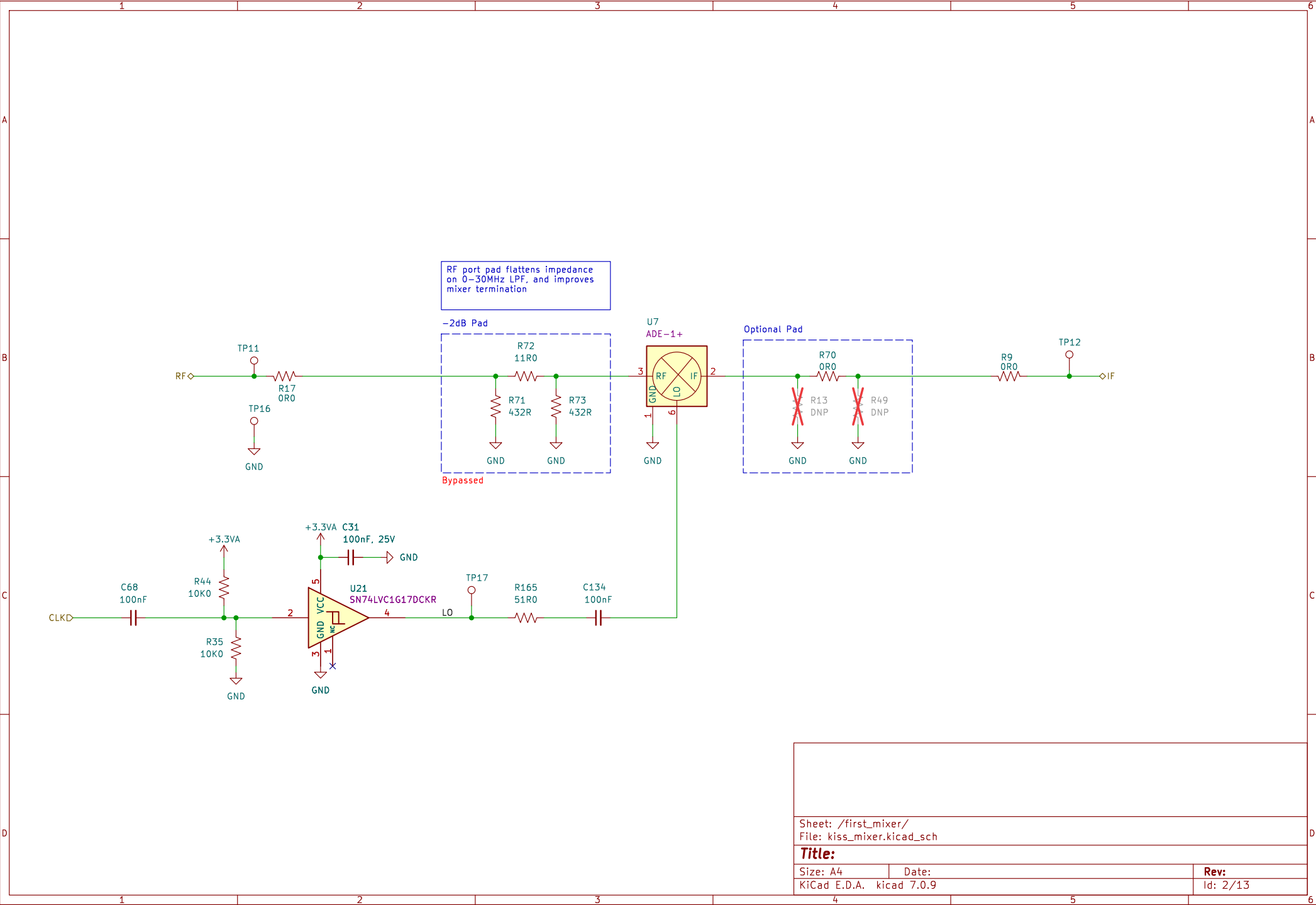






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Size: A4

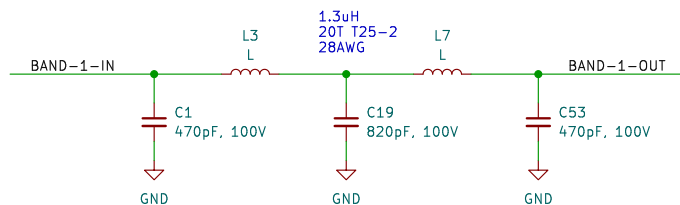
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KiCad E.D.A. kicad 7.0.9

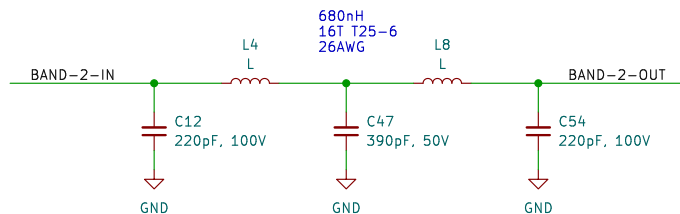
Rev:

Id: 2/13

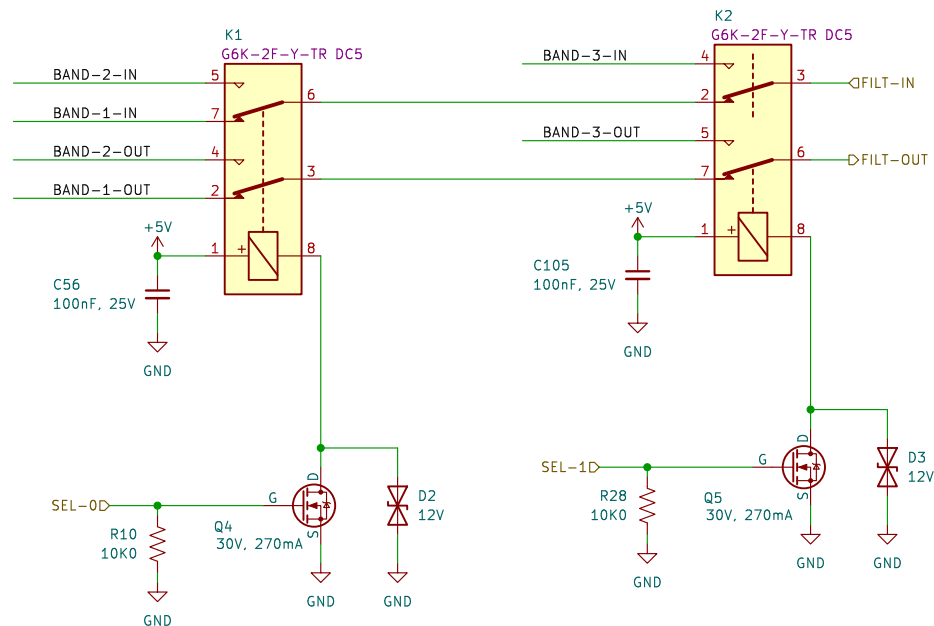
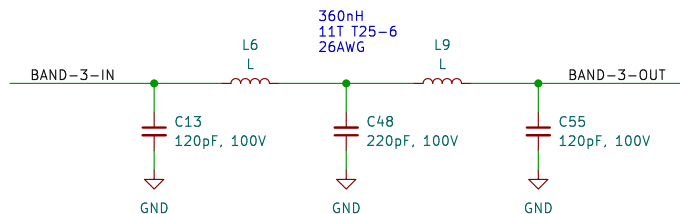
40m
8MHz 5th Order Chebyshev



30m / 20m
16MHz 5th Order Chebyshev



17m / 15m / 10m
29.5MHz 5th Order Chebyshev

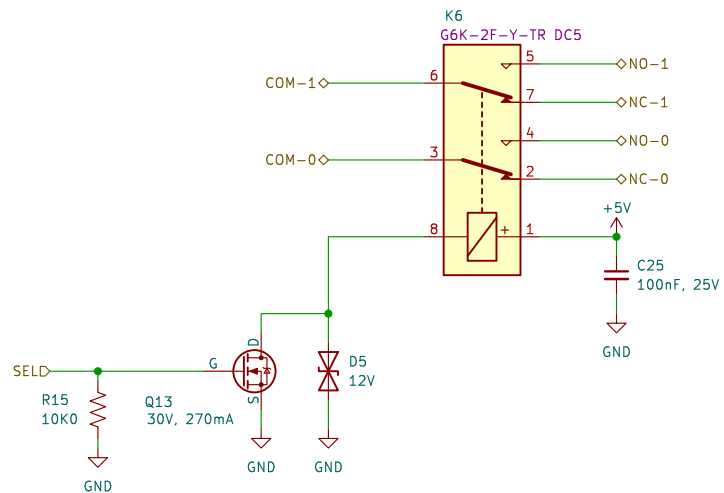


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Size: A4 Date:
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Rev:
Id: 3/13



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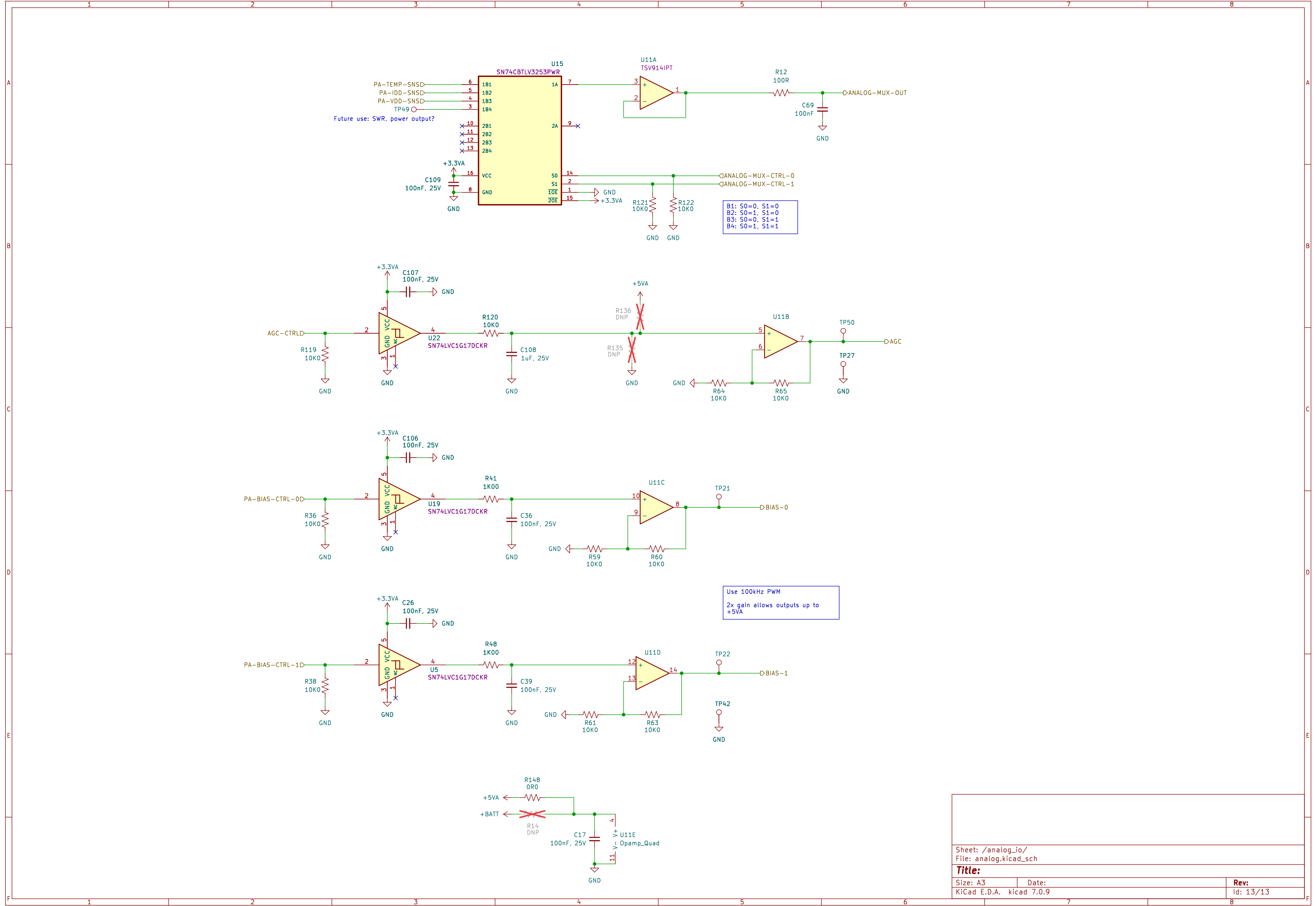
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Rev:

Id: 4/13

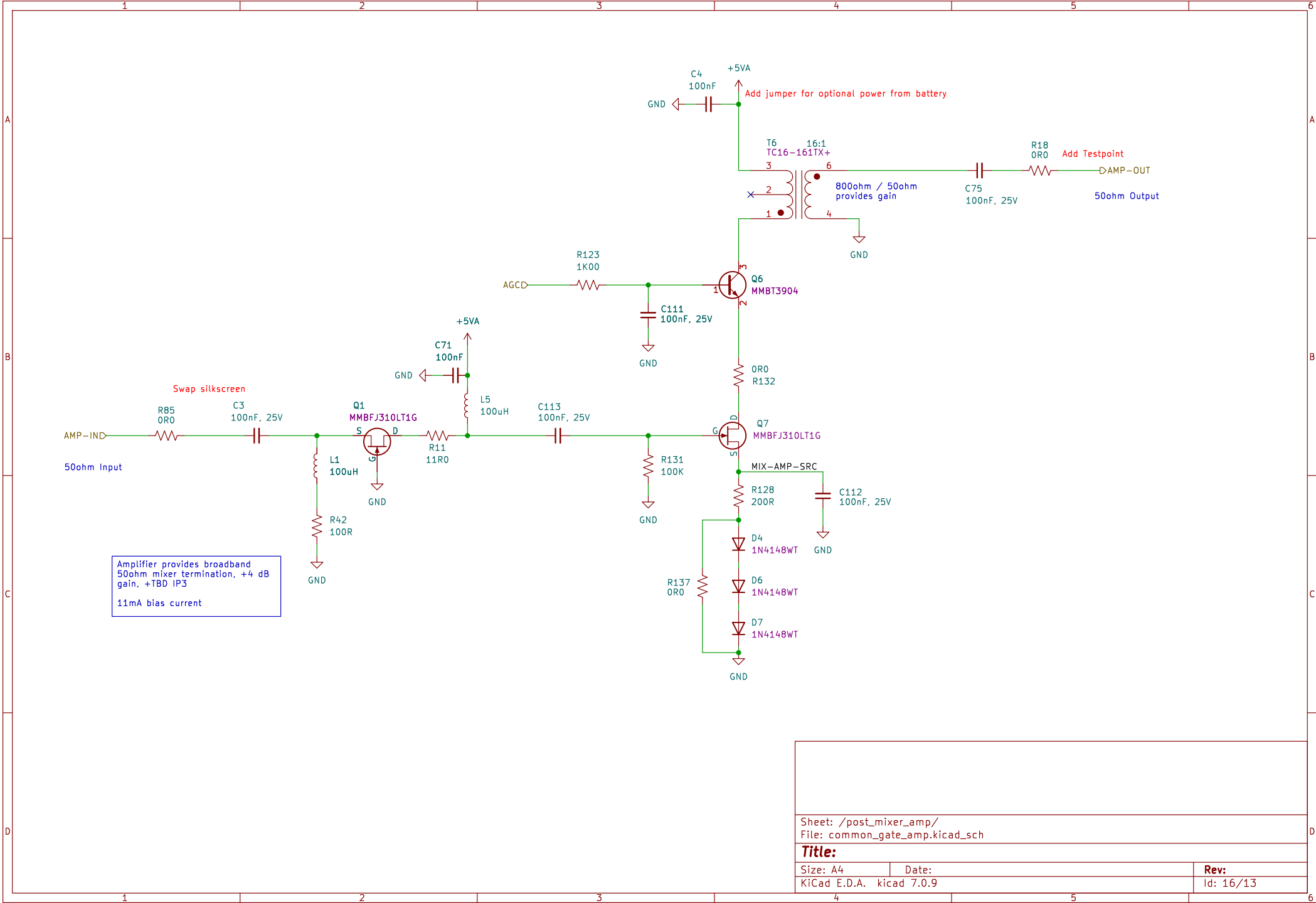


From: https://www.wb5rvz.com/sdr/RATX_V6_3/12_paf.htm

Sheet: /pa/
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Title:

Size: A2 Date: Rev:
KiCad E.D.A. kicad 7.0.9 Id: 15/13



Sheet: /post_mixer_amp/
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Title:

Size: A4

Date:

KiCad E.D.A. kicad 7.0.9

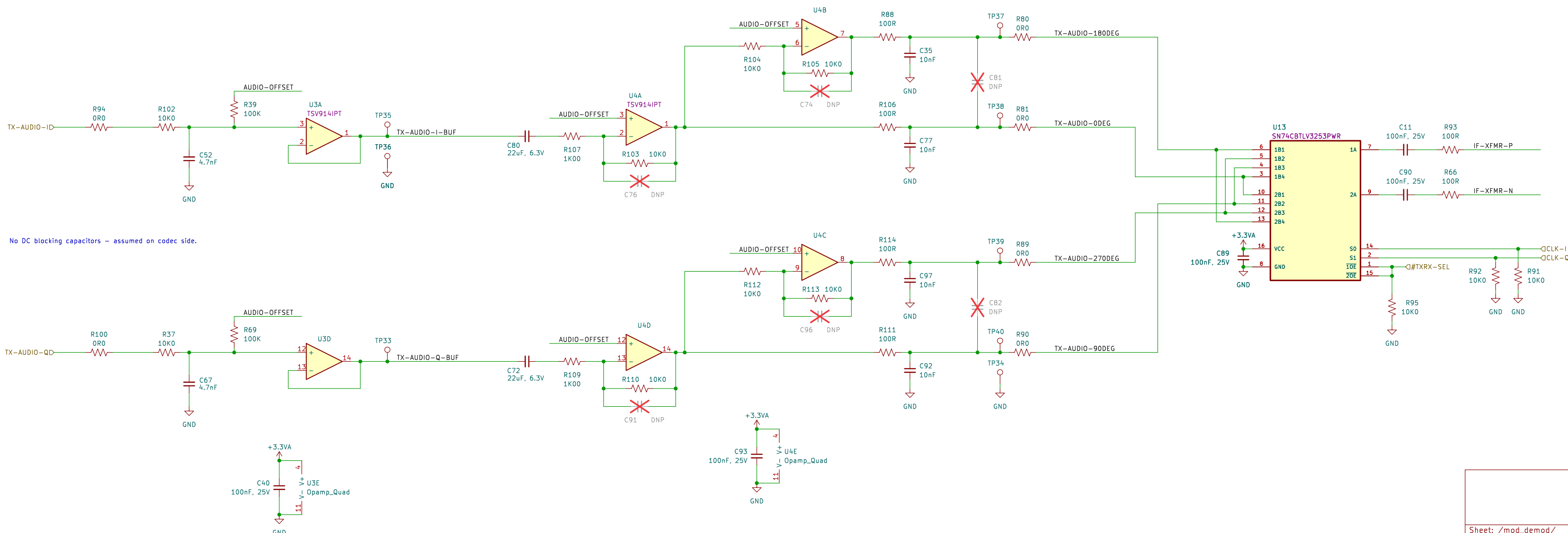
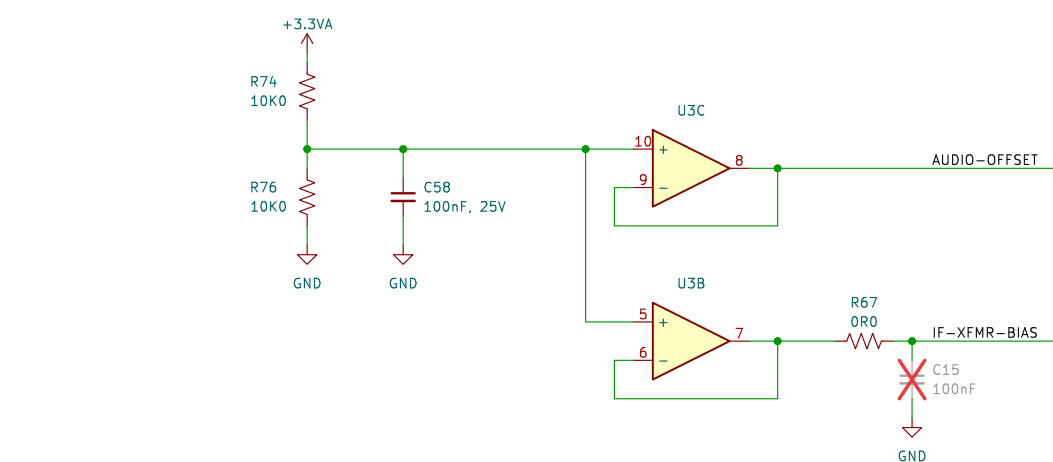
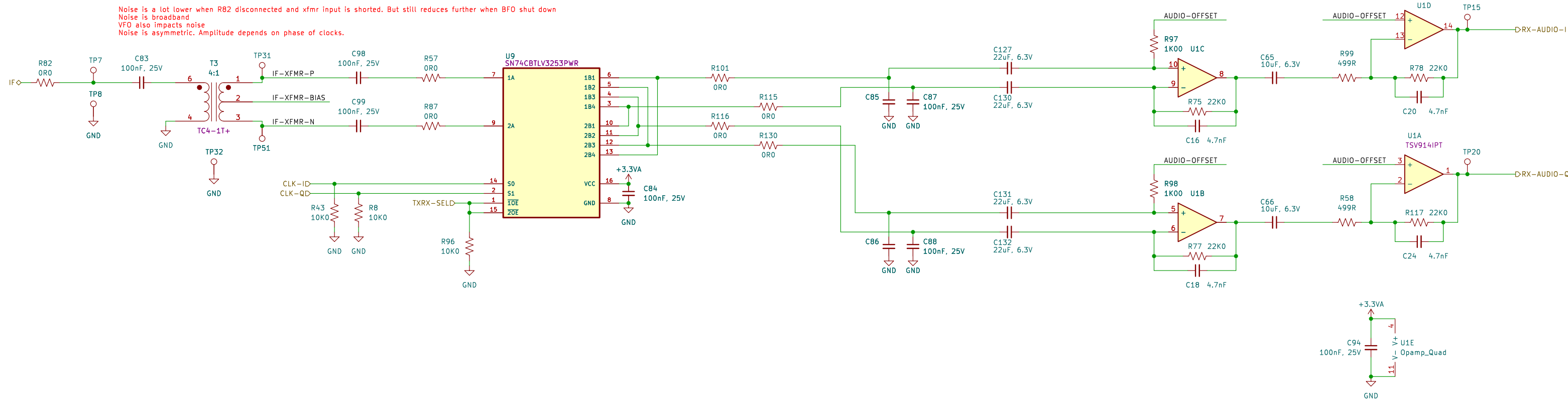
Rev:

Id: 16/13

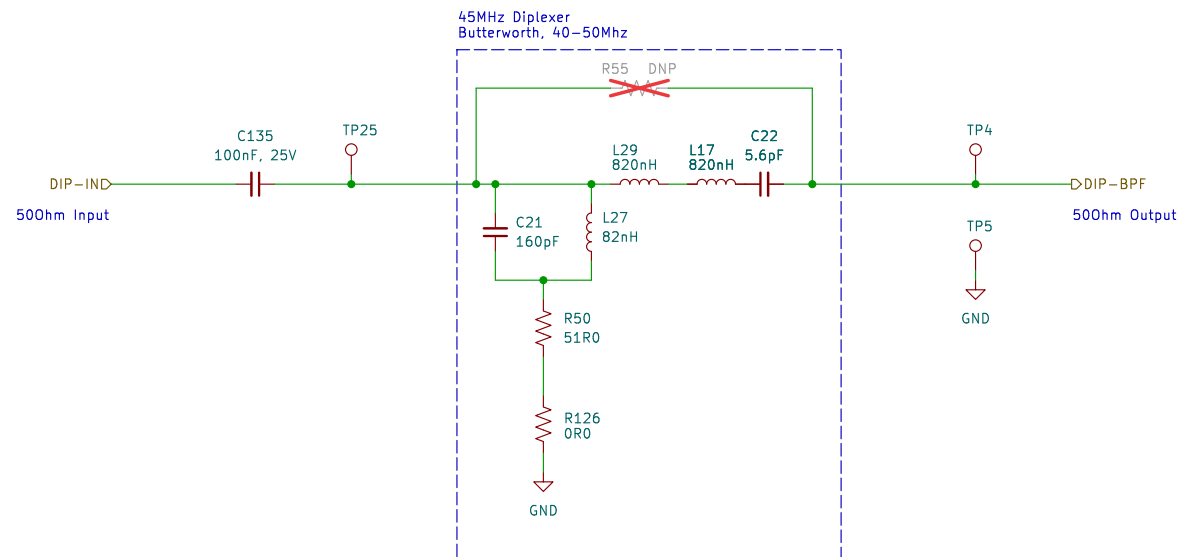
Circuit references:
https://wb5rvz.com/sdr/RXTX62/VIII_mixertx.htm
<https://qrp-labs.com/images/receiver/receiver3.s.pdf>
https://qrp-labs.com/images/qmx/manuals/schematics_rev1_a.pdf
<http://www.arri.org/files/Info/Technology/tis/info/pdf/020708qex013.pdf>
<http://www.arri.org/files/Info/Technology/tis/info/pdf/030304qex020.pdf>
<https://www.waveguide.se/?article=transceiver-build-part-4>

Tested with 2.2nF. Probably better with 1nF.

- Noise is a lot lower when R82 disconnected and xfmr input is shorted. But still reduces further when BFO shut down
- Noise is broadband
- VFO also impacts noise
- Noise is asymmetric. Amplitude depends on phase of clocks.



No DC blocking capacitors – assumed on codec side.



Even though post-mixer amplifier should present a good match across all frequencies, diplexer reduces out-of-band signals hitting the amplifier.

Possible future experiment: connect directly to xtal filter to see if post-mixer amp is needed. Maybe not?
https://www.robkalmeyer.nl/techniek/electronica/radiotechniek/hambladen/qex/2001/05_06/page45/index.html

Sheet: /diplexer/
 File: diplexer.kicad_sch

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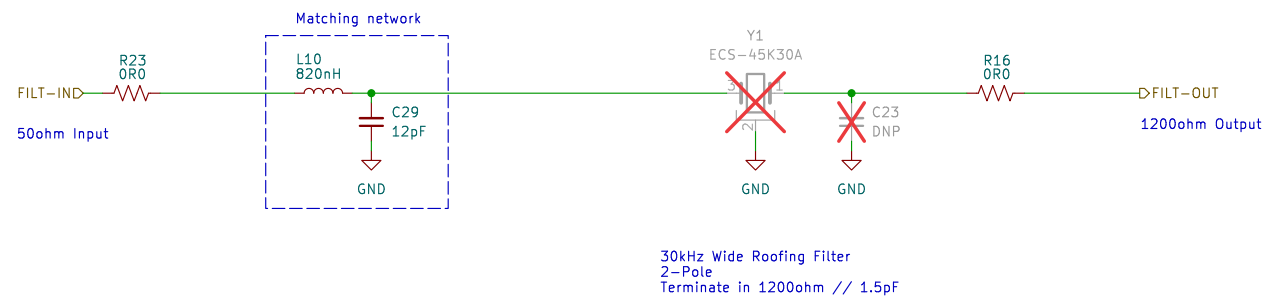
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Rev:

Id: 19/13



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File: xtal_filter.kicad_sch

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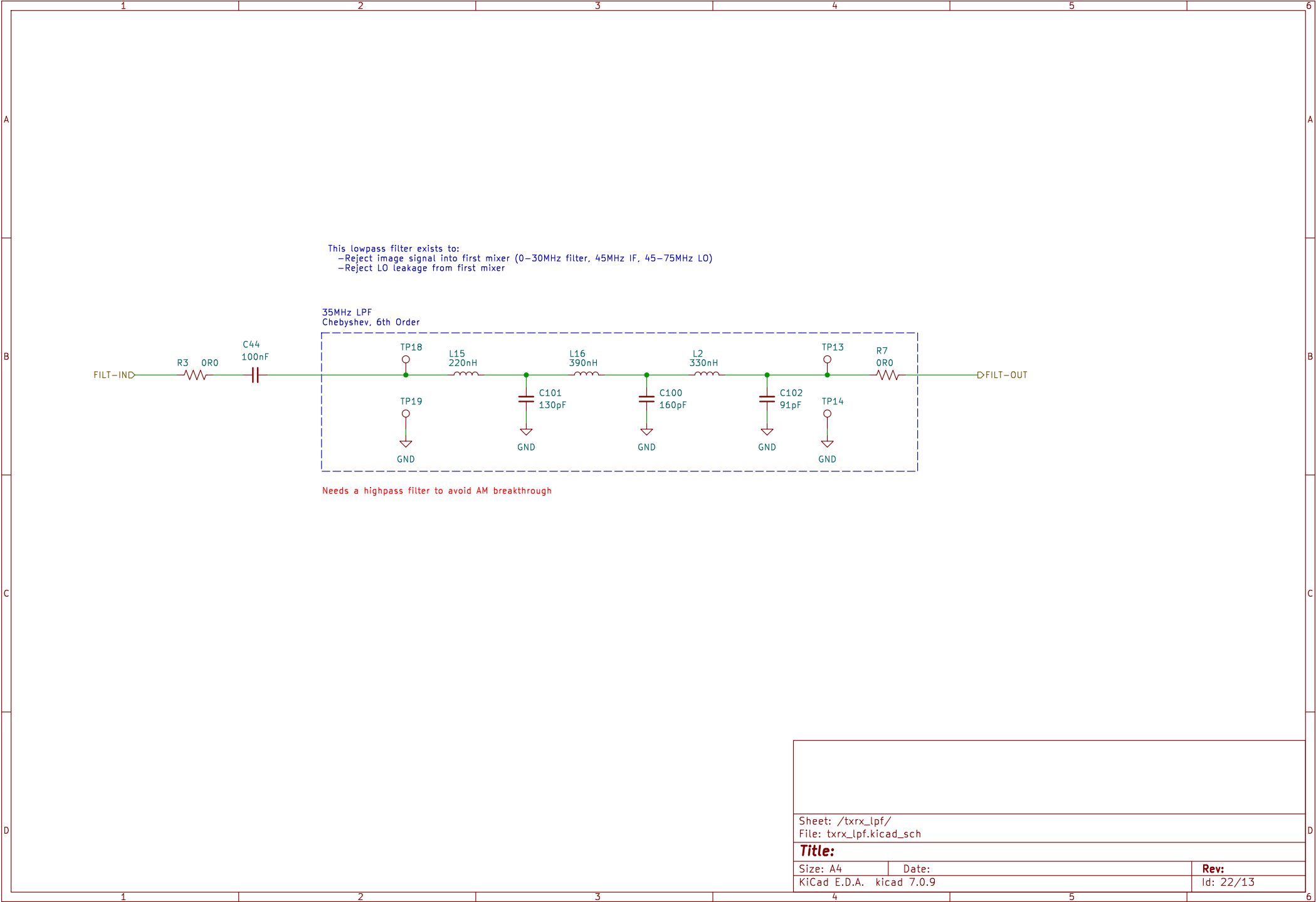
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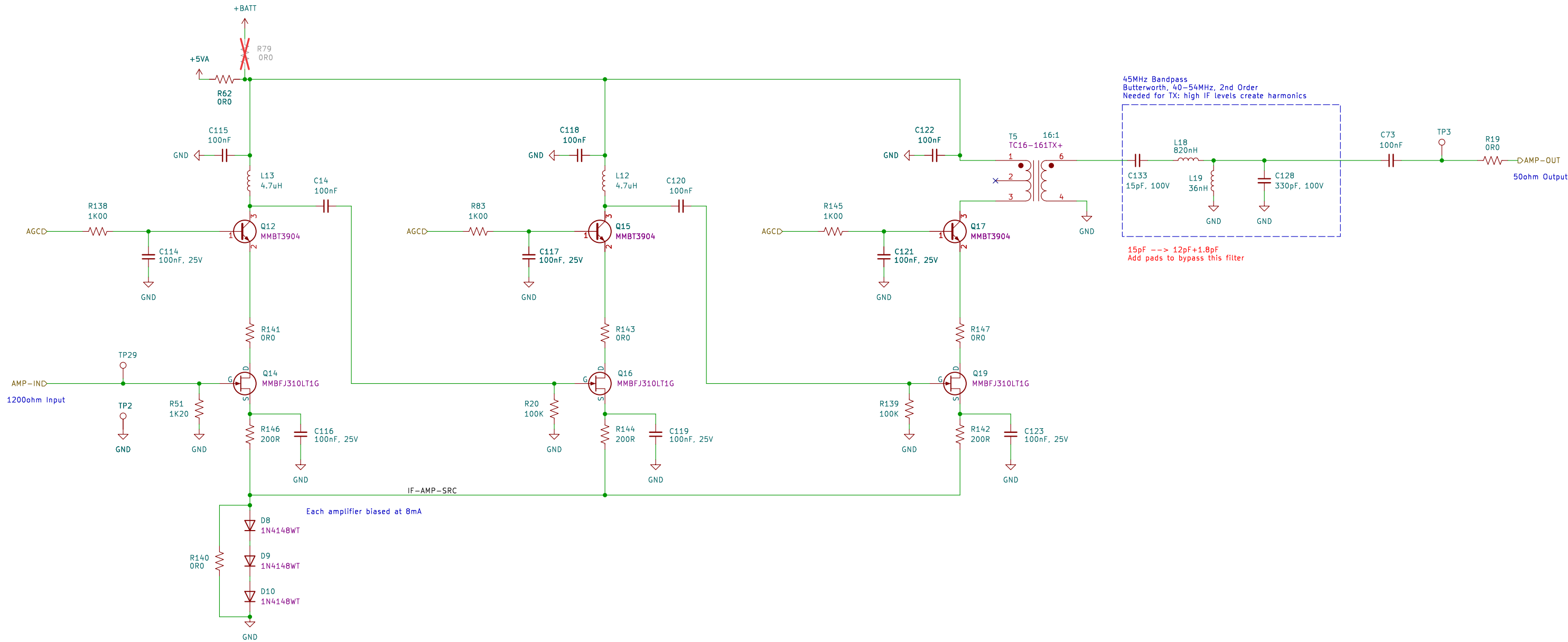
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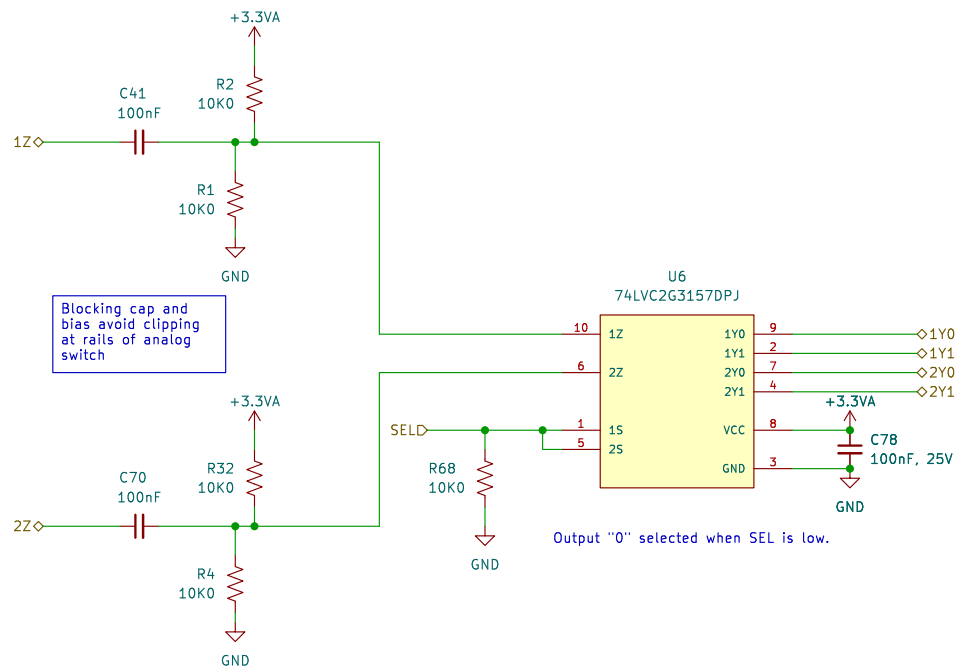
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Rev:

Id: 20/13







Sheet: /if_reverse/
File: dpdt_rf_switch.kicad_sch

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Size: A4

Date:

KiCad E.D.A. kicad 7.0.9

Rev:

Id: 24/13